

Using VMware to set up Guest Operating Systems

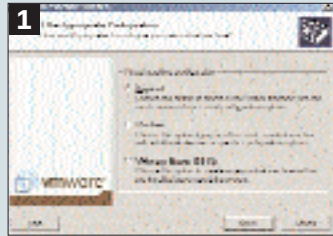
Install VMware Workstation 3.0 on your computer and give the registration number.

Click on the New Virtual Machine icon in the main window and choose the Typical option to create a new virtual machine with automatic configuration. Click on Next and select the OS you want to install as the guest OS.

Let's select Linux as the guest OS for now. Click on Next, give a name to the virtual machine and browse to select the location where you want to create the virtual disk. Click on Next to bring up the Select Network Connection dialog box. Of the four options, only the last three are relevant if



Select the guest OS



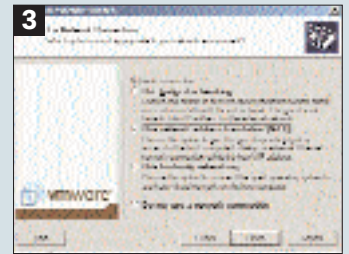
Select 'Typical' to create a new virtual machine

you are setting up your PC at home. The Use network address translation (NAT) option allows your virtual machine to access your Internet connection directly without you having to set it up. You can create a virtual network within your PC with the Use host-only network option. If you don't want any

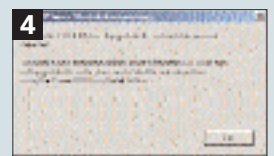
kind of network set up for the virtual machine, select the Do not use a network connection option. Click on Finish. You will be able to see the details of your virtual machine now.

Now insert a bootable CD or floppy disk to install Linux. Click on the Power On button to start the virtual machine. Linux will be installed in your virtual machine. Make sure to also install the development tools that come with your Linux installation.

One of the best programmer tools available for Linux is GCC (GNU C Compiler), which comes as part of the developer tools package. You can use GCC to program using C or C++. Code all your programs on a good editor program and run the files. Some of the good editors available for Linux are jEdit, Jext, Scintilla, Glimmer, etc. Ensure that you select the Use the VMWare Power Off button to switch off your virtual machine when you are through with your day's work.



Select the appropriate network connection



Insert a bootable CD or floppy to install the guest OS

developers site. The next thing to have is PRC-Tools, which is a GCC- (GNU C Compiler) based compiler for development on the Palm platform. (See box, 'GCC Tools for Palm'). Some of the C-C++ tools available for development on the Palm OS are Visual Form Designer and Teenie. If you are willing to pay, get Metrowerks CodeWarrior, which is much better with visual tools. You also need a (free) Palm emulator

Programming for the Palm

PRC-Tools is a compiler tools package, which helps you build applications for the Palm in C or C++. This package consists of patched versions of GCC (GNU C Compiler), binutils, GDB and post-linker tools. If you are using Windows as the platform on which you want to use PRC-tools, you will need to install Cygwin b20.1, PRC-Tools 2.1, pilRC for Win 32, Palm OS SDK, and a Palm OS emulator. If you are a Linux user, you will need GCC 2.95, PRC Tools 2.0, pilRC, Palm OS SDK, and the Palm OS emulator. You can get all these from the Palm OS site—both for Windows and Linux users.

and a Palm ROM dump used by the emulator as otherwise you won't be able to test any of the code you develop.

To develop for the Pocket PC, you would need Microsoft Visual eMbedded Tools 3.0, which consists of eMbedded Visual Basic, eMbedded Visual C++, and SDKs for Pocket PC.

Let's have a look at how we can use the emulator to test a program for Pocket PC—say, using eMbedded Visual Basic. When the application is ready to be tested, select Run in the project menu. This will compile the program and run it in the emulator.

If you want to transfer the application to a Pocket PC and run it, select the Pocket PC (Default Device) option in the drop-down menu as shown. This will run the application in your Pocket PC device.



Event emulation in eMbedded Visual Basic

Yet another tool you may require is the Java 2 Platform Micro Edition 2 (J2ME), Wireless Toolkit. This toolkit helps you develop

applications for mobile phones, two-way pagers and PDAs and works on the PocketPC and the Palm as well. You need to have Java 2 runtime environment installed on your PC to use J2ME. You can also use the free Waba programming platform.

There! That ought to start you going with your development work. Add more tools as you need. With this set up running, you'll be able to develop and test applications for a host of platforms. Get, set and go ahead on the path to *code Nirvana*. ■

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<http://www.palmos.com/dev>: Development tools for the Palm
<http://sourceforge.net/projects/prc-tools/>: PRC development tools
www.microsoft.com/mobile/: Emulators and development tools for work on mobile platforms
<http://java.sun.com/>: Toolkits for Java
www.wabasoft.com: Get the Waba program from here